

blockMeshDict (Mesh A)

```
// Mesh A Generation
// Parameters:
// D (diameter) = 1.000e+00
// Lf (fore) = 4.000e+00
// Lw (wake) = 8.000e+00
// R (outer) = 1.000e+00
// H (top/bottom) = 4.000e+00

FoamFile
{
    version 2.0;
    format ascii;
    class dictionary;
    object blockMeshDict;
}

convertToMeters 1.0;

vertices
(
    (5.0000000000000e-01 0.000000e+00 -5.0000000000000e-02) // 0
    (3.5355339059327e-01 3.535534e-01 -5.0000000000000e-02) // 1
    (3.0616169978684e-17 5.000000e-01 -5.0000000000000e-02) // 2
    (-3.5355339059327e-01 3.535534e-01 -5.0000000000000e-02) // 3
    (-5.0000000000000e-01 6.123234e-17 -5.0000000000000e-02) // 4
    (-3.5355339059327e-01 -3.535534e-01 -5.0000000000000e-02) // 5
    (-9.1848509936051e-17 -5.000000e-01 -5.0000000000000e-02) // 6
    (3.5355339059327e-01 -3.535534e-01 -5.0000000000000e-02) // 7
    (1.0000000000000e+00 0.000000e+00 -5.0000000000000e-02) // 8
    (7.0710678118655e-01 7.071068e-01 -5.0000000000000e-02) // 9
    (6.1232339957368e-17 1.000000e+00 -5.0000000000000e-02) // 10
    (-7.0710678118655e-01 7.071068e-01 -5.0000000000000e-02) // 11
    (-1.0000000000000e+00 1.224647e-16 -5.0000000000000e-02) // 12
    (-7.0710678118655e-01 -7.071068e-01 -5.0000000000000e-02) // 13
    (-1.8369701987210e-16 -1.000000e+00 -5.0000000000000e-02) // 14
    (7.0710678118655e-01 -7.071068e-01 -5.0000000000000e-02) // 15
    (8.0000000000000e+00 0.000000e+00 -5.0000000000000e-02) // 16
    (8.0000000000000e+00 7.071068e-01 -5.0000000000000e-02) // 17
    (8.0000000000000e+00 4.000000e+00 -5.0000000000000e-02) // 18
)
```

(7.0710678118655e-01 4.000000e+00 -5.000000000000e-02) // 19
(0.000000000000e+00 4.000000e+00 -5.000000000000e-02) // 20
(-7.0710678118655e-01 4.000000e+00 -5.000000000000e-02) // 21
(-4.000000000000e+00 4.000000e+00 -5.000000000000e-02) // 22
(-4.000000000000e+00 7.071068e-01 -5.000000000000e-02) // 23
(-4.000000000000e+00 0.000000e+00 -5.000000000000e-02) // 24
(-4.000000000000e+00 -7.071068e-01 -5.000000000000e-02) // 25
(-4.000000000000e+00 -4.000000e+00 -5.000000000000e-02) // 26
(-7.0710678118655e-01 -4.000000e+00 -5.000000000000e-02) // 27
(-0.000000000000e+00 -4.000000e+00 -5.000000000000e-02) // 28
(7.0710678118655e-01 -4.000000e+00 -5.000000000000e-02) // 29
(8.000000000000e+00 -4.000000e+00 -5.000000000000e-02) // 30
(8.000000000000e+00 -7.071068e-01 -5.000000000000e-02) // 31
(5.000000000000e-01 0.000000e+00 5.000000000000e-02) // 32
(3.5355339059327e-01 3.535534e-01 5.000000000000e-02) // 33
(3.0616169978684e-17 5.000000e-01 5.000000000000e-02) // 34
(-3.5355339059327e-01 3.535534e-01 5.000000000000e-02) // 35
(-5.000000000000e-01 6.123234e-17 5.000000000000e-02) // 36
(-3.5355339059327e-01 -3.535534e-01 5.000000000000e-02) // 37
(-9.1848509936051e-17 -5.000000e-01 5.000000000000e-02) // 38
(3.5355339059327e-01 -3.535534e-01 5.000000000000e-02) // 39
(1.000000000000e+00 0.000000e+00 5.000000000000e-02) // 40
(7.0710678118655e-01 7.071068e-01 5.000000000000e-02) // 41
(6.1232339957368e-17 1.000000e+00 5.000000000000e-02) // 42
(-7.0710678118655e-01 7.071068e-01 5.000000000000e-02) // 43
(-1.000000000000e+00 1.224647e-16 5.000000000000e-02) // 44
(-7.0710678118655e-01 -7.071068e-01 5.000000000000e-02) // 45
(-1.8369701987210e-16 -1.000000e+00 5.000000000000e-02) // 46
(7.0710678118655e-01 -7.071068e-01 5.000000000000e-02) // 47
(8.000000000000e+00 0.000000e+00 5.000000000000e-02) // 48
(8.000000000000e+00 7.071068e-01 5.000000000000e-02) // 49
(8.000000000000e+00 4.000000e+00 5.000000000000e-02) // 50
(7.0710678118655e-01 4.000000e+00 5.000000000000e-02) // 51
(0.000000000000e+00 4.000000e+00 5.000000000000e-02) // 52
(-7.0710678118655e-01 4.000000e+00 5.000000000000e-02) // 53
(-4.000000000000e+00 4.000000e+00 5.000000000000e-02) // 54
(-4.000000000000e+00 7.071068e-01 5.000000000000e-02) // 55
(-4.000000000000e+00 0.000000e+00 5.000000000000e-02) // 56
(-4.000000000000e+00 -7.071068e-01 5.000000000000e-02) // 57
(-4.000000000000e+00 -4.000000e+00 5.000000000000e-02) // 58
(-7.0710678118655e-01 -4.000000e+00 5.000000000000e-02) // 59
(-0.000000000000e+00 -4.000000e+00 5.000000000000e-02) // 60
(7.0710678118655e-01 -4.000000e+00 5.000000000000e-02) // 61

```

        (8.000000000000e+00 -4.000000e+00 5.000000000000e-02) // 62
        (8.000000000000e+00 -7.071068e-01 5.000000000000e-02) // 63
    );

blocks
(
    // block 0
    hex (0 8 9 1 32 40 41 33) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 1
    hex (1 9 10 2 33 41 42 34) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 2
    hex (2 10 11 3 34 42 43 35) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 3
    hex (3 11 12 4 35 43 44 36) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 4
    hex (4 12 13 5 36 44 45 37) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 5
    hex (5 13 14 6 37 45 46 38) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 6
    hex (6 14 15 7 38 46 47 39) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 7
    hex (7 15 8 0 39 47 40 32) (10 20 1) simpleGrading (2.0000e+00
1.0000e+00 1.0e+00)

    // block 8
    hex (8 16 17 9 40 48 49 41) (30 20 1) simpleGrading (4.0000e+00
1.0000e+00 1.0e+00)

    // block 9

```

```
hex (9 17 18 19 41 49 50 51) (30 30 1) simpleGrading
(4.0000e+00 4.0000e+00 1.0e+00)

// block 10
hex (10 9 19 20 42 41 51 52) (20 30 1) simpleGrading
(1.0000e+00 4.0000e+00 1.0e+00)

// block 11
hex (11 10 20 21 43 42 52 53) (20 30 1) simpleGrading
(1.0000e+00 4.0000e+00 1.0e+00)

// block 12
hex (23 11 21 22 55 43 53 54) (15 30 1) simpleGrading
(2.5000e-01 4.0000e+00 1.0e+00)

// block 13
hex (24 12 11 23 56 44 43 55) (15 20 1) simpleGrading
(2.5000e-01 1.0000e+00 1.0e+00)

// block 14
hex (25 13 12 24 57 45 44 56) (15 20 1) simpleGrading
(2.5000e-01 1.0000e+00 1.0e+00)

// block 15
hex (26 27 13 25 58 59 45 57) (15 30 1) simpleGrading
(2.5000e-01 2.5000e-01 1.0e+00)

// block 16
hex (27 28 14 13 59 60 46 45) (20 30 1) simpleGrading
(1.0000e+00 2.5000e-01 1.0e+00)

// block 17
hex (28 29 15 14 60 61 47 46) (20 30 1) simpleGrading
(1.0000e+00 2.5000e-01 1.0e+00)

// block 18
hex (29 30 31 15 61 62 63 47) (30 30 1) simpleGrading
(4.0000e+00 2.5000e-01 1.0e+00)

// block 19
hex (15 31 16 8 47 63 48 40) (30 20 1) simpleGrading
(4.0000e+00 1.0000e+00 1.0e+00)
```

```
);
```

```
edges
```

```
(  
  arc 0 1 ( 4.61940e-01 1.91342e-01 -5.00000e-02)  
  arc 1 2 ( 1.91342e-01 4.61940e-01 -5.00000e-02)  
  arc 2 3 ( -1.91342e-01 4.61940e-01 -5.00000e-02)  
  arc 3 4 ( -4.61940e-01 1.91342e-01 -5.00000e-02)  
  arc 4 5 ( -4.61940e-01 -1.91342e-01 -5.00000e-02)  
  arc 5 6 ( -1.91342e-01 -4.61940e-01 -5.00000e-02)  
  arc 6 7 ( 1.91342e-01 -4.61940e-01 -5.00000e-02)  
  arc 7 0 ( 4.61940e-01 -1.91342e-01 -5.00000e-02)  
  arc 8 9 ( 9.23880e-01 3.82683e-01 -5.00000e-02)  
  arc 9 10 ( 3.82683e-01 9.23880e-01 -5.00000e-02)  
  arc 10 11 ( -3.82683e-01 9.23880e-01 -5.00000e-02)  
  arc 11 12 ( -9.23880e-01 3.82683e-01 -5.00000e-02)  
  arc 12 13 ( -9.23880e-01 -3.82683e-01 -5.00000e-02)  
  arc 13 14 ( -3.82683e-01 -9.23880e-01 -5.00000e-02)  
  arc 14 15 ( 3.82683e-01 -9.23880e-01 -5.00000e-02)  
  arc 15 8 ( 9.23880e-01 -3.82683e-01 -5.00000e-02)  
  arc 32 33 ( 4.61940e-01 1.91342e-01 5.00000e-02)  
  arc 33 34 ( 1.91342e-01 4.61940e-01 5.00000e-02)  
  arc 34 35 ( -1.91342e-01 4.61940e-01 5.00000e-02)  
  arc 35 36 ( -4.61940e-01 1.91342e-01 5.00000e-02)  
  arc 36 37 ( -4.61940e-01 -1.91342e-01 5.00000e-02)  
  arc 37 38 ( -1.91342e-01 -4.61940e-01 5.00000e-02)  
  arc 38 39 ( 1.91342e-01 -4.61940e-01 5.00000e-02)  
  arc 39 32 ( 4.61940e-01 -1.91342e-01 5.00000e-02)  
  arc 40 41 ( 9.23880e-01 3.82683e-01 5.00000e-02)  
  arc 41 42 ( 3.82683e-01 9.23880e-01 5.00000e-02)  
  arc 42 43 ( -3.82683e-01 9.23880e-01 5.00000e-02)  
  arc 43 44 ( -9.23880e-01 3.82683e-01 5.00000e-02)  
  arc 44 45 ( -9.23880e-01 -3.82683e-01 5.00000e-02)  
  arc 45 46 ( -3.82683e-01 -9.23880e-01 5.00000e-02)  
  arc 46 47 ( 3.82683e-01 -9.23880e-01 5.00000e-02)  
  arc 47 40 ( 9.23880e-01 -3.82683e-01 5.00000e-02)  
);
```

```
boundary
```

```
(  
  
  inlet  
  {
```

```
    type patch;
    faces
    (
      (22 54 55 23)
      (23 55 56 24)
      (24 56 57 25)
      (25 57 58 26)
    );
  }

  outlet
  {
    type patch;
    faces
    (
      (18 50 49 17)
      (17 49 48 16)
      (16 48 63 31)
      (31 63 62 30)
    );
  }

  cylinder
  {
    type wall;
    faces
    (
      (0 32 33 1)
      (1 33 34 2)
      (2 34 35 3)
      (3 35 36 4)
      (4 36 37 5)
      (5 37 38 6)
      (6 38 39 7)
      (7 39 32 0)
    );
  }

  top
  {
    type symmetryPlane;
    faces
    (
```

```
        (22 54 53 21)
        (21 53 52 20)
        (20 52 51 19)
        (19 51 50 18)
    );
}

bottom
{
    type symmetryPlane;
    faces
    (
        (26 58 59 27)
        (27 59 60 28)
        (28 60 61 29)
        (29 61 62 30)
    );
}

);

{
```

log.blockMesh (Mesh A)

```
/*-----
--*\
=====
\\    /  F i e l d      | OpenFOAM: The Open Source CFD Toolbox
\\    /  O peration    | Website:  https://openfoam.org
\\    /  A nd          | Version:  7
\\\/    M anipulation   |
\*-----
--*/
Build   : 7
Exec    : blockMesh
Date    : Mar 06 2025
Time    : 18:33:05
Host    : "c211-027.frontera.tacc.utexas.edu"
PID     : 206787
I/O     : uncollated
Case    :
/work2/10531/andressuniaga/frontera/COE-347-SP25/run/circular-cylinder
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using
timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// * * * * *
* //
Create time

Creating block mesh from

"/work2/10531/andressuniaga/frontera/COE-347-SP25/run/circular-cylinder/sys
tem/blockMeshDict"
Creating block edges
No non-planar block faces defined
Creating topology blocks
Creating topology patches

Creating block mesh topology
--> FOAM Warning :
    From function Foam::polyMesh::polyMesh(const Foam::IOobject &
```



```
Foam::Field<Foam::Vector<double>> &&, const Foam::List<Foam::cellShape> &,
const Foam::List<Foam::List<Foam::face>> &, const Foam::List<Foam::word> &,
const Foam::PtrList<Foam::dictionary> &, const Foam::word &, const
Foam::word &, bool)
    in file meshes/polyMesh/polyMeshFromShapeMesh.C at line 873
    Found 40 undefined faces in mesh; adding to default patch.
```

Check topology

```
Basic statistics
Number of internal faces : 28
Number of boundary faces : 64
Number of defined boundary faces : 64
Number of undefined boundary faces : 0
Checking patch -> block consistency
```

Creating block offsets
Creating merge list .

Creating polyMesh from blockMesh

Creating points with scale 1

```
Block 0 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 1 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 2 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 3 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 4 cell size :
i : 0.034505 .. 0.0690099
```

```
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 5 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 6 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 7 cell size :
i : 0.034505 .. 0.0690099
j : 0.0196337 .. 0.0196337 0.0209886 .. 0.0209886 0.0196337 ..
0.0196337 0.0209886 .. 0.0209886
k : 0.1
Block 8 cell size :
i : 0.107248 .. 0.428993 0.10726 .. 0.42904 0.107248 .. 0.428993
0.10726 .. 0.42904
j : 0.0392674 .. 0.0392674 0.0392074 .. 0.039058 0.0392674 ..
0.0392674 0.0392074 .. 0.039058
k : 0.1
Block 9 cell size :
i : 0.111736 .. 0.446943
j : 0.050451 .. 0.201804
k : 0.1
Block 10 cell size :
i : 0.0392674 .. 0.0392674 0.0392074 .. 0.039058 0.0392674 ..
0.0392674 0.0392074 .. 0.039058
j : 0.0459635 .. 0.183854 0.0459753 .. 0.183901 0.0459635 .. 0.183854
0.0459753 .. 0.183901
k : 0.1
Block 11 cell size :
i : 0.0392674 .. 0.0392674 0.039058 .. 0.0392074 0.0392674 ..
0.0392674 0.039058 .. 0.0392074
j : 0.050451 .. 0.201804 0.0500341 .. 0.200136 0.050451 .. 0.201804
0.0500341 .. 0.200136
k : 0.1
Block 12 cell size :
i : 0.401311 .. 0.100328
j : 0.050451 .. 0.201804
```

```
k : 0.1
Block 13 cell size :
i : 0.365616 .. 0.0914039 0.36571 .. 0.0914275 0.365616 .. 0.0914039
0.36571 .. 0.0914275
j : 0.0353553 .. 0.0353553 0.0358313 .. 0.0346553 0.0353553 ..
0.0353553 0.0358313 .. 0.0346553
k : 0.1
Block 14 cell size :
i : 0.401311 .. 0.100328 0.397995 .. 0.0994988 0.401311 .. 0.100328
0.397995 .. 0.0994988
j : 0.0353553 .. 0.0353553 0.0346553 .. 0.0358313 0.0353553 ..
0.0353553 0.0346553 .. 0.0358313
k : 0.1
Block 15 cell size :
i : 0.401311 .. 0.100328
j : 0.201804 .. 0.050451
k : 0.1
Block 16 cell size :
i : 0.0353553 0.0349631 .. 0.0355947 0.0353553 0.0349631 .. 0.0355947
j : 0.201804 .. 0.050451 0.200136 .. 0.0500341 0.201804 .. 0.050451
0.200136 .. 0.0500341
k : 0.1
Block 17 cell size :
i : 0.0353553 .. 0.0353553 0.0355947 .. 0.0349631 0.0353553 ..
0.0353553 0.0355947 .. 0.0349631
j : 0.183854 .. 0.0459635 0.183901 .. 0.0459753 0.183854 .. 0.0459635
0.183901 .. 0.0459753
k : 0.1
Block 18 cell size :
i : 0.111736 .. 0.446943
j : 0.201804 .. 0.050451
k : 0.1
Block 19 cell size :
i : 0.111736 .. 0.446943 0.111319 .. 0.445275 0.111736 .. 0.446943
0.111319 .. 0.445275
j : 0.0392674 .. 0.0392674 0.039058 .. 0.0392074 0.0392674 ..
0.0392674 0.039058 .. 0.0392074
k : 0.1
Creating cells
Creating patches
```

There are no merge patch pairs edges

Writing polyMesh

Mesh Information

boundingBox: (-4 -4 -0.05) (8 4 0.05)

nPoints: 17530

nCells: 8500

nFaces: 34265

nInternalFaces: 16735

Patches

patch 0 (start: 16735 size: 100) name: inlet

patch 1 (start: 16835 size: 100) name: outlet

patch 2 (start: 16935 size: 160) name: cylinder

patch 3 (start: 17095 size: 85) name: top

patch 4 (start: 17180 size: 85) name: bottom

patch 5 (start: 17265 size: 17000) name: defaultFaces

End

log.checkMesh (Mesh A)

```
/*-----
--*\
=====
\\      /  F i e l d      | OpenFOAM: The Open Source CFD Toolbox
\\      /  O peration     | Website:  https://openfoam.org
\\      /  A nd           | Version:  7
\\\/     M anipulation    |
\*-----
--*/
Build   : 7
Exec    : checkMesh
Date    : Mar 06 2025
Time    : 18:33:09
Host    : "c211-027.frontera.tacc.utexas.edu"
PID     : 206900
I/O     : uncollated
Case    :
/work2/10531/andressuniaga/frontera/COE-347-SP25/run/circular-cylinder
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using
timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// * * * * *
* //
Create time

Create polyMesh for time = 0

Time = 0

Mesh stats
  points:          17530
  internal points:  0
  faces:           34265
  internal faces:  16735
  cells:           8500
  faces per cell:  6
  boundary patches: 6
```

```
point zones:      0
face zones:       0
cell zones:       0
```

Overall number of cells of each type:

```
hexahedra: 8500
prisms:    0
wedges:    0
pyramids:  0
tet wedges: 0
tetrahedra: 0
polyhedra: 0
```

Checking topology...

```
Boundary definition OK.
Cell to face addressing OK.
Point usage OK.
Upper triangular ordering OK.
Face vertices OK.
Number of regions: 1 (OK).
```

Checking patch topology for multiply connected surfaces...

Patch	Faces	Points	Surface topology
inlet	100	202	ok (non-closed singly connected)
outlet	100	202	ok (non-closed singly connected)
cylinder	160	320	ok (non-closed singly connected)
top	85	172	ok (non-closed singly connected)
bottom	85	172	ok (non-closed singly connected)
defaultFaces	17000	17530	ok (non-closed singly connected)

Checking geometry...

```
Overall domain bounding box (-4 -4 -0.05) (8 4 0.05)
Mesh has 2 geometric (non-empty/wedge) directions (1 1 0)
Mesh has 2 solution (non-empty) directions (1 1 0)
All edges aligned with or perpendicular to non-empty directions.
Boundary openness (-6.97029e-19 -1.23283e-17 -3.12937e-16) OK.
Max cell openness = 2.18444e-16 OK.
Max aspect ratio = 12.6414 OK.
Minimum face area = 0.0007007. Maximum face area = 0.0901947. Face
area magnitudes OK.
Min volume = 7.007e-05. Max volume = 0.00901947. Total volume =
9.52148. Cell volumes OK.
Mesh non-orthogonality Max: 43.3859 average: 12.6085
```

Non-orthogonality check OK.

Face pyramids OK.

Max skewness = 0.903887 OK.

Coupled point location match (average 0) OK.

Mesh OK.

End